

Quite simply put KRO is the ability of a keyboard to distinguish all the keys that are simultaneously pressed. The notation is preceded by a number which tells you how many keys can be pressed at the same time. This way 3KRO means that 3 keys can be pressed simultaneously and the keyboard will recognize each key press. There are keyboards that come with 6KRO and nKRO (any number of keys may be simultaneously pressed) they are generally a bit expensive given that this functionality is present in most high end keyboards. PS/2 keyboards are theoretically capable of figuring out each and every key pressed but that is when each key is wired separately. This is not the case with keyboards as most keyboards use two plastic sheets with circuits etched over them with mutually isolated contact points that are very close to each other. These contact points register a keypress when the space between them is closed by the key.

Ghosting

The above image shows a typical keyboard membrane, you can see how most of the contact lines are placed in a grid like pattern. Each key is not separately connected but rather a combination of horizontal row and vertical columns are used to figure out which key was pressed. Now imagine if three keys were to be pressed in the same row. Only one row would light up while three vertical columns will light up. This tells the keyboard circuitry that three keys have been pressed. Now if 4 keys were to be pressed and that too 2 keys in one row and the 2 immediate keys right above them. This square pattern now registers 2 rows and 2 columns. Again the computer knows which keys were pressed. Now finally remove one of the keys pressed in the previous combination. This still triggers 2 rows and 2 columns. So even when you've pressed 3 keys the computer is unsure whether it was 3 or 4. This phenomenon of detecting a key that hasn't been pressed is called Ghosting. The whole issue has less to do with PS/2 or USB but rather the way the membrane is designed.

Layouts

The keyboard is broken down into certain groups. Most of the alphabets form a huge chunk of the keys and are aptly called the character keys. Then there are Modifier keys which consist of Shift, Alt and Ctrl which we use to give commands of access Menu bars. Then come the navigation keys which allow the user to move about the workspace. The number keys exist in two places, the first is right above the character keys for ease